Northwestern University, Department of Neurology; Robert J. Havey, MD Institute for Global Health

Clinical Utility and Feasibility of Smartphone-based EEG Across Hospitals in Kenya

Nomin Enkhtsetseg[1], William Lehn-Schiøler[2-3], Anton Storgaard[2], Magnus Guldberg Pedersen[2], Dylan R. Rice[1], George Wambugu[2], Nshimiyimana Jules Fidele[4], Anca Alina Arbune[5], Sidsel Armand Larsen[6], Sandor Beniczky[6-8], Farrah J. Mateen[1]

[1] Davee Department of Neurology, Feinberg School of Medicine, Northwestern University, Chicago, IL, USA; [2] BrainCapture, Kongens Lyngby, Denmark; [3] Department of Health Technology, Technical University of Denmark, Kongens Lyngby, Denmark; [4] Kenyatta University Teaching, Referral and Research Hospital, Nairobi, Kenya; [5] Department of Neurology, Institutul Clinic Fundeni, Bucharest, Romania; [6] Danish Epilepsy Center, Filadelfia, Denmark; [7] Copenhagen University Hospital Rigshospitalet, Copenhagen, Denmark; [8] Department of Clinical Medicine, University of Copenhagen, Copenhagen, Denmark

Northwestern Medicine® Feinberg School of Medicine

Background

- Epilepsy affects >50 million people globally, with >80% of the burden in low-and middle-income countries (LMICs).[1-3]
- Access to electroencephalography (EEG) remains limited due to cost, infrastructure, workforce constrains.[3]
- These gaps contribute to delayed diagnoses and preventable morbidity.[3]
- In Kenya, the burden is further amplified by:
  - Increased exposure to preventable neurological risk factors (e.g., traumatic brain injury, severe malaria, perinatal complications, infections)[4-5]
  - Limited diagnostic infrastructure[3]
  - Transportation and financial barriers[3]
- The BrainCapture BC-1 smartphone-based EEG system offers a scalable solution to expand neurodiagnostic access, aligning with priorities outlined in the World Health Organization Intersectoral Global Action Plan (WHO IGAP).[6-7]

Objective

To report findings from >2000 point-of-care smartphone-based EEG recordings, assessing the feasibility, clinical utility, and diagnostic yield of this approach in routine clinical care across Kenya.

Methods

Retrospective study across 20 hospitals and imaging clinics in Kenya.

- **EEG Acquisition:** Standardized 30-minute, 27-lead, asleep and awake scalp EEGs were performed on patients of all ages undergoing routine clinical EEGs, usually for suspected epilepsy. EEGs were acquired by trained healthcare workers.
- **EEG Interpretation:** Recordings were remotely reviewed and remotely analyzed by certified, trained neurophysiologists for (1) quality of recording, (2) presence of epileptiform discharges, and (3) other abnormal EEG findings.
- **Outcomes:**
- **Clinical utility** was evaluated by the ability of smartphone EEG to provide actionable diagnostic information (e.g., epileptiform activity) that supports clinical decision-making;
- **Feasibility** was assessed by interpretability rate; and
- **Diagnostic yield** was determined by the proportion of abnormal and epileptiform findings.

Figure 1. Clinical indications for smartphone EEG within study cohort

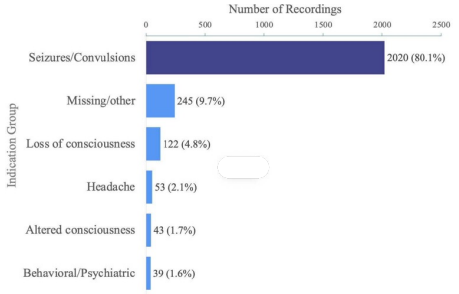


Figure 2. Age-specific distribution of smartphone EEG findings

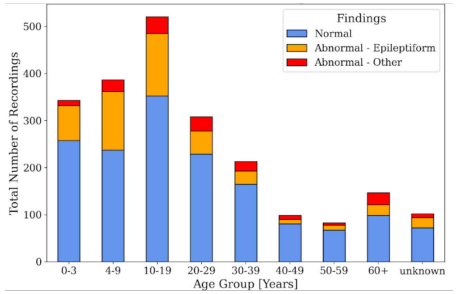


Table 1. Demographic and clinical characteristics of the study cohort

Study Sample	
Total EEG recordings analyzed	2,522 (89.4)
Excluded recordings	300 (10.6)
Sex	
Male	1488 (59)
Female	1034 (41)
Age Group (years)	
0–3	340 (13.5)
4–9	385 (15.3%)
10–19	520 (20.6%)
20–29	310 (12.3%)
30–39	215 (8.5%)
40–49	95 (3.8%)
50–59	80 (3.2%)
≥60	145 (5.7%)
Unknown	100 (4.0%)
Key Medical History	
Head trauma	292 (11.6)
Family history of seizures	269 (10.7)
Stroke/Heart attack	61 (2.4)
Brain surgery	28 (1.1)
EEG Findings	
Normal	1,783 (70.7)
Epileptiform abnormalities	550 (21.8)
Non-epileptiform abnormalities	189 (7.5)

Results

A total of 2,522 participants **(Table 1)** underwent smartphone EEGs. Primary clinical indications for EEG **(Figure 1)** were: seizures/ convulsions (80%), altered/loss of consciousness (7%), headache (2%), behavioral/psychiatric (2%), and other suspected seizure-related events or unknown indications (10%).

Feasibility:

- **89.4% clinically interpretable** recordings
- 10.6% uninterpretable (artifact, impedance, or duration)
- Median interpretation time to read an EEG recording: 102 minutes

Diagnostic Yield and Clinical Utility:

- 29% abnormal EEGs (of which **74% were epileptiform, 26% non-epileptiform**)

Age-Specific Patterns **(Figure 2)**:

- Epileptiform abnormalities were more common in younger individuals, whereas older adults more frequently exhibited non-epileptiform abnormalities.

Limitations

This study was observational and non-population-based, which may limit generalizability. There was no follow-up data which prevented assessment of downstream clinical outcomes.

Conclusions

Smartphone EEG demonstrated high interpretability (89.4%) and clinically meaningful diagnostic yield. Findings support feasibility of decentralized EEG in resource-limited settings.

Future research should explore integration of AI-assisted EEG interpretation to enhance scalability and reduce reliance on specialist readers in LMIC settings. Furthermore, smartphone-based EEG has the potential to improve diagnostic yield, inform clinical decision-making, and address existing gaps in the epidemiology of global epilepsy. These approaches may also support reverse innovation, with applications in underserved populations in high-income countries.

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Study Contact: Nomin Enkhtsetseg, BA; nomin.enkhtsetseg@northwestern.edu